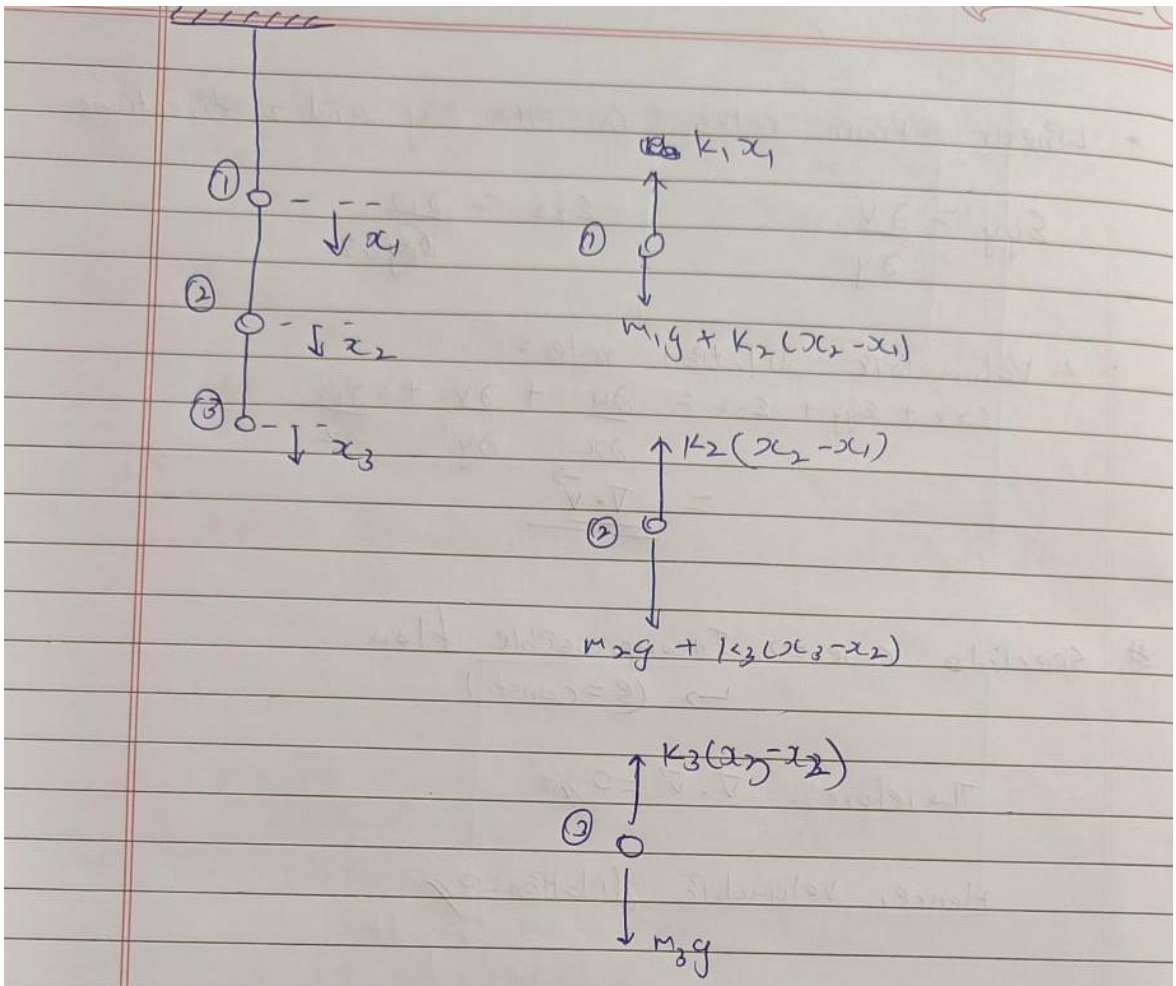
LAB 3: Systems of Linear Algebraic Equations

Objective: To represent a system of linear algebraic equations in matrix form and obtain the solution.

<u>Question 1:</u> Bungee Jumper



$$m_1 \frac{d^2 x_1}{dt^2} = m_1 g + k_2(x_2 - x_1) - k_1 x_1$$

$$m_2 \frac{d^2 x_2}{dt^2} = m_2 g + k_3(x_3 - x_2) + k_2(x_1 - x_2)$$

$$m_3 \frac{d^2 x_3}{dt^3} = m_3 g + k_3(x_2 - x_3)$$

$$\therefore \begin{aligned} (k_1 + k_2)x_1 - k_2 x_2 + 0 &= m_1 g \\ -k_2 x_1 + (k_2 + k_3)x_2 - k_3 x_3 &= m_2 g \\ 0 - k_3 x_2 + k_3 x_3 &= m_3 g \end{aligned}$$

```
clear
clc
k = [50 100 50];
m = [60 70 80];

mg = [m(1); m(2); m(3)] .* 9.81;

k = [(k(1)+k(2)) -k(2) 0;
    -k(2) k(2)+k(3) -k(3);
    0 -k(3) k(3)];

x = inv(k)*mg;
disp('The value of x is: ')
disp(x)

xi = [20;40;60];
xf = x + xi;
disp('The value of x_final is')
disp(xf)
```

Command Window

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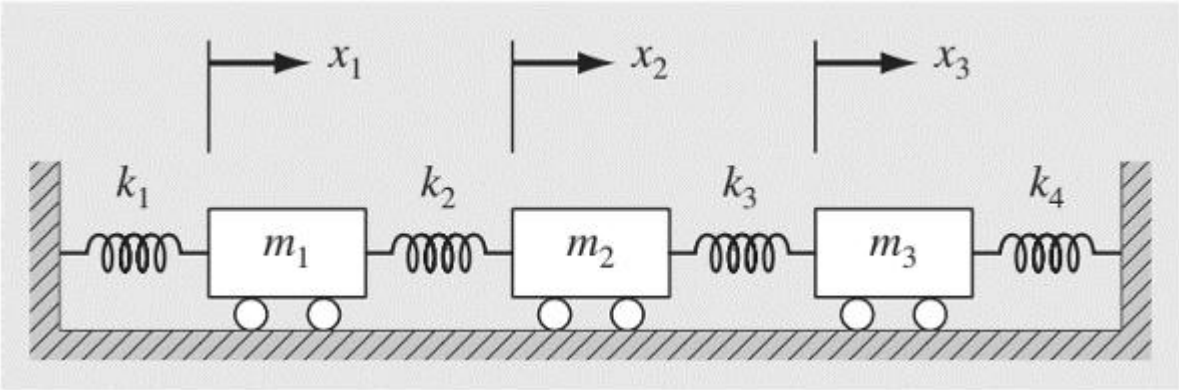
The value of x is:
   41.2020
   55.9170
   71.6130

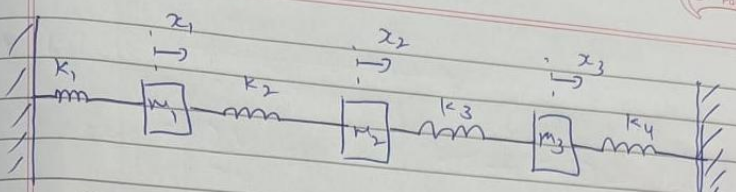
The value of x_final is
   61.2020
   95.9170
  131.6130
```

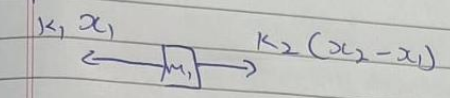
Question 2: Consider the three mass-four spring system in Fig. Determining the equations of motion from ΣFx =max for each mass using its free-body diagram results in the

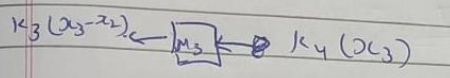


Derive the governing equation for acceleration for all the three blocks?

Assuming : $x_3 > x_2 > x_1$







$$k_1 x_1 - k_2(x_2 - x_1) = m_1 a_1$$
$$k_2(x_2 - x_1) - k_3(x_3 - x_2) = m_2 a_2$$
$$k_3(x_3 - x_2) + k_4(x_3) = m_3 a_3$$

$$(k_1 + k_2)x_1 - k_2 x_2 = m_1 a_1$$
$$(k_2 + k_3)x_2 - k_2 x_1 - k_3 x_3 = m_2 a_3$$
$$(k_3 + k_4)x_3 - k_3 x_2 = m_3 a_3$$

$$\begin{bmatrix} k_1 + k_2 & -k_2 & 0 \\ -k_2 & k_2 + k_3 & -k_3 \\ 0 & -k_3 & k_3 + k_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} m_1 a_1 \\ m_2 a_2 \\ m_3 a_3 \end{bmatrix}$$

$$\begin{bmatrix} (k_1 + k_2)/m_1 & -(k_2/m_1) & 0 \\ (-k_2)/m_2 & (k_2 + k_3)/m_2 & -(k_3/m_2) \\ 0 & -(k_3/m_3) & (k_3 + k_4)/m_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

```
clc; clear;
k1= 10;
k4 = 10;
k2 = 30;
k3=30;
m1 = 1;
m2 = 1;
m3=1;

x1=0.05;
x2=0.04;
x3=0.03;

k = [(k1+k2) -(k2) 0; -k2 k2+k3 -k3; 0 -k3 k3+k4];
m = [m1;m2;m3];

k_m = k./m;

x = [x1;x2;x3];

a = - (k_m * x);

disp(a)
```

Command Window

```
   -0.8000
         0
         0
```